

# Online Regret Minimization via Rademacher Complexities

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## 1 Introduction

In the setting of online learning, a learner plays real-valued actions against an adversary, who responds by playing an outcome after seeing the learner's action. The learner then incurs some loss dictated by a known loss function at each iteration. This iterative process can be modeled as a zero-sum game between learner and adversary, where the goal of the player is to implement a (potentially nondeterministic) strategy that approaches the best strategy in hindsight, despite only observing the sequence of data/outcomes thus far. Put simply, the difficulty in developing efficient strategies for the learner arises in the inability to make predictions conditioned on future outcome(s).

Informally, the *regret* of the player is the difference between observed loss and the loss it could have incurred if it knew the adversary's strategy (also potentially nondeterministic) in advance. Extensive work has shown that, even when no assumptions are made on the strategy of the adversary, many loss settings permit asymptotic convergence to "no regret" of some form [1]. This result has strong implications for most topics involving probabilistic forecasting, such as weather forecasting, quantile estimation, supervised learning, and hedging, to name a few.

The online learning framework also lies in the broader field of statistical learning theory, which seeks to abstract the results of particular well-performing algorithms into general statements about the conditions necessary to permit efficient no-regret algorithms. In this work, we investigate a computationally efficient framework for learners to minimize their regret. While a naive strategy for the learner may be to iterate through all exponentially many action-outcome transcripts, picking the one yielding the optimal regret, [2] shows that we may frame the expected regret as a more computationally feasible stochastic process instead. This allows us to use the framework established in [3] to compute actions

for the learner that obtain near-optimal regret, differing only by a constant factor at worst.

The work is structured as follows: we begin by defining the setting and outlining some related literature. Next, we define the sequential Rademacher complexity of a hypothesis class and its implications for convergence guarantees, briefly discussing the more generalized definition of empirical processes. We then use *conditional* sequential Rademacher complexities to show how one may derive computationally efficient algorithms, with a notable example being the exponential weights algorithm.

## 1.1 Setting

The online learning setting can be treated as a game between a learner and an adversary, where the game lasts  $T$  steps. Both the learner and adversary have public action spaces  $\mathcal{F}, \mathcal{Y} \subseteq \mathbb{R}^d$ . In this work, we'll assume  $d = 1$ , so all actions by the learner/adversary are real-valued. At each iteration, the learner plays a prediction  $f_t \in \mathcal{F}$ , to which the adversary may respond with outcome  $y_t \in \mathcal{Y}$ , and the learner incurs loss  $\ell(f_t, y_t)$  for a known loss function  $\ell : \mathcal{F} \times \mathcal{Y} \rightarrow \mathbb{R}$ . We assume that sufficient conditions on  $\ell, \mathcal{F}, \mathcal{Y}$  are placed for von Neumann's classic minimax theorem to hold - see [4]. Namely, when  $\mathcal{F}, \mathcal{Y}$  are compact and convex, and  $\ell$  is concave in its first argument & convex in its second, the optimal value of the game doesn't depend on who plays first.

Both the learner and adversary are free to play randomized strategies, i.e. the learner may play a "randomized prediction" which isn't realized until after the adversary chooses a (potentially randomized) outcome. For sake of clutter, we will freely interchange the notation for deterministic predictions  $f \in \mathcal{F}$  and randomized predictions  $f \in \Delta\mathcal{F}$ , and similarly for outcomes.

While the adversary is assumed to be nearly omnipotent and may see the learner's present/past predictions, the learner only knows the outcomes seen thus far. Define the transcript of the game  $\pi_T := \pi_{1:T} = \{(f_t, y_t)\}_{t \in [T]}$ . The goal of the learner is to minimize its expected *external regret*, or its incremental loss relative to the best *fixed* action in hindsight:

$$\text{Reg}(\pi_T) = \mathbb{E}_{f,y} \left[ \sum_{t=1}^T \ell(f_t, y_t) - \inf_{f \in \mathcal{F}} \sum_{t=1}^T \ell(f, y_t) \right]$$

The naive brute-force strategy for the learner is to iterate through all possible strategies to find the one that minimizes their regret:

$$f_t = \arg \min_{\Delta\mathcal{F}} \left\{ \sup_{y_t} \inf_{f_{t+1}} \sup_{y_{t+1}} \dots \sup_{y_T} \mathbb{E} \left[ \sum_{s=t}^T \ell(f_s, y_s) - \inf_{f \in \mathcal{F}} \sum_{s=1}^T \ell(f, y_s) \right] \right\}$$

Such a strategy would obtain precisely the minimax value of the game:

$$\mathcal{V}(\mathcal{F}) = \inf_{f_1 \in \mathcal{F}} \sup_{y_1 \in \mathcal{Y}} \dots \inf_{f_T \in \mathcal{F}} \sup_{y_T \in \mathcal{Y}} \mathbb{E}[\text{Reg}(\pi_T)]$$

In the toy problem of predicting an adversary’s coin flips, the minimax regret is  $O(\sqrt{T})$ , or average regret  $\frac{O(\sqrt{T})}{T} = O(\frac{1}{\sqrt{T}})$ , which rapidly diminishes to 0. This is as any learner strategy obtains expected loss  $T/2$  while the best option between heads/tails in hindsight obtains loss  $T/2 - \Theta(\sqrt{T})$  as per the central limit theorem. More broadly, our goal is to obtain diminishing average regret for any game, while also being able to compute learner strategies in an efficient manner.

## 1.2 Related Work

The online learning framework has developed rapidly in recent years, with wide ranging applications in the realms of binary prediction, linear regression [5], and classification [6]. Each of these fields harbor efficient algorithms to compute optimal strategies, such as follow the perturbed leader for sequential linear regression over a bounded space. Binary classification in particular also has close connections to algorithmic fairness, most famously the notion of *calibration*: algorithms that make predictions that are unbiased, both in general as well as conditioned on its own predictions ([7], [8]).

Quantifying the complexity of a hypothesis class began with the seminal work on VC dimension [9], which was used to explain the amount of data needed to converge to the best hypothesis in the class, i.e. a ”probably approximately correct” hypothesis (for more info on PAC learning, see [10]). Understanding how close one can get to the best hypothesis in a class has had renowned interest in the realm of online supervised learning, which also sparked alternative measures of complexity, including Gaussian Complexity, maximum discrepancy, integrated complexity, and of course the Rademacher complexity which we study here ([11], [12], [2]).

## 2 Rademacher Complexities & Convergence

We first consider the online *supervised* learning setting, in which the learner’s action space is a class of ”hypothesis” functions  $\mathcal{F} \subseteq \mathcal{Y}^{\mathcal{X}}$  for some context space  $\mathcal{X}$ . The goal of the learner is to then predict the best hypothesis mapping the observed contexts  $\{x_1, \dots, x_T\}$  to outcomes  $\{y_1, \dots, y_T\}$ .

The maximum regret of the learner is dependent on the *complexity* of the hypothesis class: more complex hypothesis classes can be expected to have better performance, but

may be more difficult to optimize over in an efficient manner. To formalize this notion of complexity, we first introduce the notion of *trees*:

**Definition 1.** A  $\mathcal{Y}$ -valued tree of depth  $T$  is a complete rooted binary tree defined by labeling functions  $y_t : \{\pm 1\}^{t-1} \rightarrow \mathcal{Y}$ , where  $y_t$  provides the labels for the nodes at depth  $i$  in the binary tree.

Intuitively,  $\mathcal{Y}$ -valued trees represent the potential strategies arising from an observed transcript of *independent* binary outcomes: upon observing a sequence  $\epsilon_{1:t-1} = \{\epsilon_1, \dots, \epsilon_{t-1}\}$ , the strategy dictated by the  $\mathcal{Y}$ -valued tree would be  $y_t(\epsilon_{1:t-1})$ .

Continuing the toy problem of predicting coin flips, consider a variation where the learner begins with \$1, and each coin flip is a wager to gain/lose 50% of their current wealth. The binary tree containing the values corresponding to the first three iterations is shown below. At the first iteration, the learner gains  $\epsilon_1 0.5$ . An outcome of  $\epsilon_1 = -1$  corresponds to the left node, as now the learner is only eligible to win/lose 0.25, whereas  $\epsilon_1 = +1$  corresponds to the right node, where the learner may win/lose 0.75. This continues for  $\epsilon_2, \epsilon_3$ , and so on.

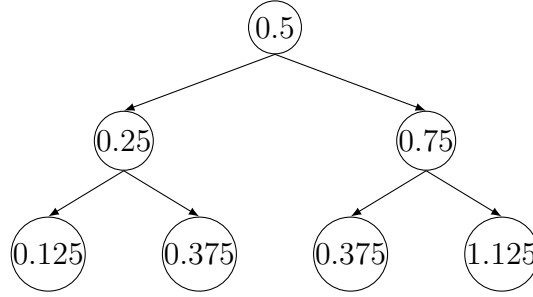


Figure 1: A  $\mathbb{R}$ -valued tree for the 50% betting game up to depth 2

In this instance, the "adversary strategy" is the fixed 50% gain/decrease in the learner's current wealth. Given this information, the learner may enact their own strategy  $f$ , which would yield corresponding action  $f(y_t(\epsilon_{1:t}))$ .

Now, recall that a Rademacher random variable  $\epsilon$  is an r.v. uniform over  $\{-1, 1\}$ . With this in mind, we define the sequential Rademacher complexity of a hypothesis class:

**Definition 2** (Sequential Rademacher complexity [2]). *The Rademacher complexity of a hypothesis class  $\mathcal{F}$  is defined as*

$$\mathfrak{R}(\mathcal{F}) = \sup_y \mathbb{E}_\epsilon \sup_{f \in \mathcal{F}} \sum_{t=1}^T \epsilon_t f(y_t(\epsilon_{1:t}))$$

where the supremum is taken over all  $\mathcal{Y}$ -valued trees  $y$ , and  $\epsilon = \{\epsilon_1, \epsilon_2, \dots, \epsilon_T\}$  is a sequence of independent Rademacher random variables.

Intuitively, the Rademacher complexity is the best expected performance of the learner  $f \in \mathcal{F}$  when pitted against the best expected strategy of the adversary  $y$ , given a sequence of independent inputs.

It remains to be shown that the Rademacher complexity can be used to bound the optimal regret of the learner's strategy, as both the learner and adversary's strategies in online learning tether around *dependent* data. A different approach capturing a similar notion of comparing the expected performance of the learner against an adversary is the more general *empirical process with dependent data*, where the inputs are no longer independent but can be assumed to be accumulated over many iterations.

**Definition 3** (Definition 7.6 of [12]). *An empirical process is a stochastic process  $\mathbb{M}_f := \mathbb{M}_f(y_{1:t})$ , indexed by learner strategy  $f \in \mathcal{F}$ , is defined as*

$$\mathbb{M}_f = \sum_{t=1}^T (\mathbb{E}[f(y_t)|y_1, \dots, y_{t-1}] - f(y_t))$$

where  $y_{1:T} = \{y_t\}_{t \in [T]}$  is a stochastic process.

As it turns out, an empirical process with dependent data can be tightly bounded by the Rademacher complexity (a lower bound can be intuited if the data points are independent, which essentially gives back the Rademacher complexity). We now state the following theorem relating the Rademacher complexity & the optimal regret of our learner:

**Theorem 2.1** (Theorem 2 of [13]). *For any empirical process  $\mathbb{M}_f := \mathbb{M}_f(y)$ ,*

$$\mathbb{E} \sup_{f \in \mathcal{F}} \mathbb{M}_f \leq 2\mathfrak{R}(\mathcal{F})$$

The above theorem is, in essence, an empirical process convergence theorem, claiming the deviation from the expected value of a game is bounded by the sequential Rademacher complexity of the given class. Note that the empirical process  $\mathbb{M}_f$  is similar in definition to the loss incurred by strategy  $f$ .

We now return to the general setting of online learning and real-valued action spaces  $\mathcal{F} \subseteq \mathbb{R}$ , where we spectacularly derive an identical upper bound on the optimal loss incurred by the learner, which is simply the minimax value of the game:

**Theorem 2.2** (Theorem 7 of [2]). *The minimax value of an online learning game is bounded by*

$$\mathcal{V}(\mathcal{F}) \leq 2\mathfrak{R}(\ell(\mathcal{F}))$$

where  $\ell(\mathcal{F}) = \{\ell(f, \cdot) : f \in \mathcal{F}\}$

This connection is somewhat surprising, as the sequential Rademacher complexity is predominantly used in the literature to quantify the complexity of function classes. But as we'll show in the following section, we can tweak the notion of a sequential Rademacher complexity to general real-valued action spaces, which will help us establish a framework for minimizing the regret.

## 3 Relaxation Algorithms

### 3.1 General Framework

We first formalize what the "remaining value" of the game is at any step.

**Definition 4.** Given transcript  $\pi_t = \{(f_s, y_s)\}_{s \in [t]}$ , the conditional value of the game is recursively defined as

$$\mathcal{V}(\mathcal{F}|y_1, \dots, y_{t-1}) = \inf_{f_t} \sup_{y_t} (\mathbb{E}[\ell(f_t, y_t)] + \mathcal{V}(\mathcal{F}|y_1, \dots, y_{t-1}, y_t))$$

where  $\mathcal{V}(\mathcal{F}|\emptyset) = \mathcal{V}(\mathcal{F})$  and  $\mathcal{V}(\mathcal{F}|y_1, \dots, y_T) = -\inf_f \sum_{t=1}^T \ell(f, y_t)$ .

At each iteration, the optimal strategy of the learner is to minimize the conditional value of the game, whether through brute force or a computationally efficient method.

As the conditional value of the game is often difficult to quantify, one may instead hope to optimize for a computationally feasible yet tight upper bound on this value.

**Definition 5.** A relaxation is a sequence of real-valued functions  $\{\text{Rel}(\mathcal{F}|y_1, \dots, y_t)\}_{t \in [T]}$ . A relaxation is admissible if, for any complete sequence of outcomes  $\{y_t\}_{t \in [T]}$  and  $t \in [T-1]$ ,

$$\text{Rel}(\mathcal{F}|y_1, \dots, y_{t-1}) \geq \inf_{f_t} \sup_{y_t} (\mathbb{E}[\ell(f_t, y_t)] + \text{Rel}(\mathcal{F}|y_1, \dots, y_{t-1}, y_t))$$

where  $\text{Rel}(\mathcal{F}|\emptyset) = \text{Rel}(\mathcal{F})$  and  $\text{Rel}(\mathcal{F}|y_1, \dots, y_T) \geq -\inf_{f \in \mathcal{F}} \sum_{t=1}^T \ell(f, y_t)$ . The learner's algorithm corresponding to this relaxation is to play the action minimizing the relaxed value at each time step:

$$f_t = \arg \min_{\Delta_{\mathcal{F}}} \sup_{y_t} \{\mathbb{E}[\ell(f_t, y_t)] + \text{Rel}(\mathcal{F}|y_1, \dots, y_t)\}$$

Any admissible relaxation upper bounds the conditional value of the game. As such, one can expect that an algorithm obtaining a small relaxed value should obtain a small conditional value at each time step as well, and hence have small overall regret.

**Lemma 3.1** (Proposition 1 in [3]). *For any algorithm arising out of admissible relaxation  $\{\text{Rel}(\mathcal{F}|y_1, \dots, y_t)\}_{t \in [T]}$  with arbitrary corresponding outcome sequence  $\{y_t\}_{t \in [T]}$ , we have that*

$$\text{Reg}(\pi_T) \leq \text{Rel}(\mathcal{F})$$

*Proof.* We observe that

$$\mathbb{E}_f \left[ \sum_{t=1}^T \ell(f_t, y_t) \right] - \inf_{f \in \mathcal{F}} \sum_{t=1}^T \ell(f, y_t) \leq \mathbb{E}_f \left[ \sum_{t=1}^T \ell(f_t, y_t) \right] + \text{Rel}(\mathcal{F}|y_1, \dots, y_t)$$

As  $\text{Rel}(\mathcal{F}|y_1, \dots, y_{T-1}) \geq \mathbb{E}[\ell(f_T, y_T)] + \text{Rel}(\mathcal{F}|y_1, \dots, y_T)$  given  $f_T$  is played according to the algorithm corresponding to the relaxation, we see

$$\text{Reg}(\pi_T) \leq \mathbb{E}_f \left[ \sum_{t=1}^{T-1} \ell(f_t, y_t) \right] + \text{Rel}(\mathcal{F}|y_1, \dots, y_{T-1})$$

Generally,  $\text{Rel}(\mathcal{F}|y_1, \dots, y_{t-1}) \geq \mathbb{E}[\ell(f_t, y_t)] + \text{Rel}(\mathcal{F}|y_1, \dots, y_t)$  for all  $t \in [T]$ , so recursively we get  $\text{Reg}(\pi_T) \leq \text{Rel}(\mathcal{F}|\emptyset) = \text{Rel}(\mathcal{F})$ . As this holds for any fixed realization of outcomes  $\{y_t\}_{t \in [T]}$ , it must also hold for any randomized sequence of outcomes, so  $\text{Reg}(\pi_T) = \mathbb{E}_{f,y} \left[ \sum_{t=1}^T \ell(f_t, y_t) - \inf_{f \in \mathcal{F}} \sum_{t=1}^T \ell(f, y_t) \right] \leq \text{Rel}(\mathcal{F})$ , as desired<sup>1</sup>.  $\square$

We conclude that it suffices for the learner to find a suitable relaxation and, at each iteration, find an action minimizing its value. However, it is still unclear what such a relaxation looks like. We now define one such relaxation:

**Definition 6.** *The conditional sequential Rademacher complexity for some outcome sequence  $\{y_t\}_{t \in [T]}$  is defined as*

$$\mathfrak{R}(\mathcal{F}|y_1, \dots, y_{t-1}) = \sup_{y'} \mathbb{E}_\epsilon \left[ \sup_{f_t \in \mathcal{F}} 2 \sum_{s=t}^T \epsilon_s \ell(f_t, y'_{s-t}(\epsilon_{t:s})) - \sum_{s=1}^{t-1} \ell(f_t, y_s) \right]$$

where  $\epsilon = \epsilon_{t:T}$  is a sequence of  $T-t$  independent Rademacher variables, and  $y'$  is a  $T-t$  depth  $\mathcal{Y}$ -valued tree.

**Lemma 3.2** (Proposition 22.2 of [12]). *For any  $\{y_t\}_{t \in [T]}$ , the sequence of conditional sequential Rademacher complexities  $\{\mathfrak{R}(\mathcal{F}|y_1, \dots, y_t)\}_{t \in [T]}$  is an admissible relaxation.*

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<sup>1</sup>In this work, we'll only bound expected regret, but so long as the loss function  $\ell$  is bounded, one can directly show that  $\text{Reg} \leq \text{Rel}$  w.h.p by the Hoeffding-Azuma inequality

*Proof.* We induct on  $1 \leq t \leq T$ , starting with  $t = T$  and decrementing  $t$ . Clearly,  $\mathfrak{R}(\mathcal{F}|y_1, \dots, y_T) \geq \sup_{f_T \in \mathcal{F}} \sum_{t=1}^T \ell(f_T, y_t)$ . Now, for general  $1 \leq t < T$ , we seek to show

$$\sup_{y_t} \inf_{f_t} \mathbb{E}_{f_t, y_t} \ell(f_t, y_t) + \mathfrak{R}(\mathcal{F}|y_1, \dots, y_t) \leq \mathfrak{R}(\mathcal{F}|y_1, \dots, y_{t-1})$$

We first assume the minimax theorem holds, which allows us to interchange the inf and sup representing whether the learner/adversary plays first. Then,

$$\inf_{f_t} \sup_{y_t} \mathbb{E}[\ell(f_t, y_t)] + \mathfrak{R}(\mathcal{F}|y_1, \dots, y_t) = \sup_{y_t} \inf_{f_t} \mathbb{E}[\ell(f_t, y_t)] + \mathfrak{R}(\mathcal{F}|y_1, \dots, y_t)$$

We now rewrite  $\mathfrak{R}(\mathcal{F}|y_1, \dots, y_t) = \sup_{y'} \mathbb{E}_{\epsilon_{t+1:T}} \sup_{f \in \mathcal{F}} K_{t+1}(f) - L_t(f)$ , where  $K_{t+1}(f) = 2 \sum_{s=t+1}^T \epsilon_s \ell(f, y'_s(\epsilon_{t+1:s}))$  represents the randomized loss to be incurred in the future, while  $L_t(f) = \sum_{s=1}^t \ell(f, y_s)$  is the loss in the past. Then, the above equals

$$\begin{aligned} & \sup_{y_t} \inf_{f_t} \mathbb{E}_{f_t, y_t} \ell(f_t, y_t) + \sup_{y'} \mathbb{E}_{\epsilon} \sup_f (K_{t+1}(f) - L_t(f)) \\ &= \sup_{y_t} \mathbb{E}_{y_t} \left[ \sup_{y'} \mathbb{E}_{\epsilon} \sup_f (K_{t+1}(f) - L_{t-1}(f)) + \inf_{f_t} \mathbb{E}_{y_t} [\ell(f_t, y_t)] - \ell(f, y_t) \right] \end{aligned}$$

For space, denote  $M = \sup_{y'} \mathbb{E}_{\epsilon} \sup_f (K_{t+1}(f) - L_{t-1}(f))$ . We upper bound the inner inf over  $f_t$  as

$$\begin{aligned} & \leq \sup_{y_t} \mathbb{E}_{y_t} [M + \mathbb{E}_{y_t} [\ell(f, y_t)] - \ell(f, y_t)] \\ &= \sup_{y_t} \mathbb{E}_{y_t^{(0)}, y_t^{(1)} \sim y_t} [M + \ell(f, y_t^{(0)}) - \ell(f, y_t^{(1)})] \end{aligned}$$

where  $y_t^{(0)}, y_t^{(1)}$  are i.i.d drawn from distribution  $y_t$ . As they are i.i.d,  $\ell(f, y_t^{(0)}) - \ell(f, y_t^{(1)})$  is symmetric, and thus equal to  $\mathbb{E}_{\epsilon_t} [\epsilon_t (\ell(f, y_t^{(0)}) - \ell(f, y_t^{(1)}))]$  for Rademacher r.v.  $\epsilon_t$ , which gives us

$$\begin{aligned} \sup_{y_t} \mathbb{E}_{y_t^{(0)}, y_t^{(1)} \sim y_t} \mathbb{E}_{\epsilon_t} [M + \epsilon_t (\ell(f, y_t^{(0)}) - \ell(f, y_t^{(1)}))] &\leq \sup_{y_t^{(0)}, y_t^{(1)}} \mathbb{E}_{\epsilon_t} [M + \epsilon_t (\ell(f, y_t^{(0)}) - \ell(f, y_t^{(1)}))] \\ &= \sup_{y_t^{(0)}} \mathbb{E}_{\epsilon_t} [M/2 + \epsilon_t \ell(f, y_t^{(0)})] + \sup_{y_t^{(1)}} \mathbb{E}_{\epsilon_t} [M/2 + \epsilon_t \ell(f, y_t^{(1)})] \\ &= \sup_{y_t} \mathbb{E}_{\epsilon_t} \left[ \sup_{y'} \mathbb{E}_{\epsilon_{t+1:T}} \sup_f (K_{t+1}(f) - L_{t-1}(f) + 2\epsilon_t \ell(f, y_t)) \right] \end{aligned}$$

Finally, we observe

$$\sup_{y_t} \mathbb{E}_{\epsilon_t} K_{t+1}(f) + 2\epsilon_t \ell(f, y_t) = \sup_{y_t} \mathbb{E}_{\epsilon_t} \sup_{y'} \mathbb{E}_{\epsilon_{t+1:T}} \sup_f 2 \sum_{s=t+1}^T \epsilon_s \ell(f, y'_s(\epsilon_{t+1:s})) + 2\epsilon_t \ell(f, y_t)$$



We may augment the depth  $T - t - 1$  binary tree of  $y'$  to a depth  $T - t$  depth tree via the outer supremum, and merge  $\epsilon_t$  to be randomized alongside the other  $\epsilon_{t+1:T}$ , which gets us that the above is equal to  $K_t(f)$ . We conclude that

$$\begin{aligned} \sup_{y_t} \mathbb{E}_{\epsilon_t} \sup_{y'} \mathbb{E}_{\epsilon_{t+1:T}} \sup_f (K_{t+1}(f) - L_{t-1}(f) + 2\epsilon_t \ell(f, y_t)) &= \sup_{y_t} \mathbb{E}_{\epsilon_t} \sup_{y'} \mathbb{E}_{\epsilon_{t+1:T}} \sup_f (K_t(f) - L_{t-1}(f)) \\ &= \mathfrak{R}(\mathcal{F} | y_1, \dots, y_{t-1}) \end{aligned}$$

as desired.  $\square$

### 3.2 Example: Exponential Weights

The exponential weights algorithm is one of the most ubiquitous algorithms for finite-class online learning. The basic premise is as follows: suppose you have  $k$  financial experts predicting whether tomorrow's market will be up/down, corresponding to binary values 0/1. Each day, you also predict 0 or 1, and incur some loss if your prediction is wrong. You don't know which of the experts is the most reliable, so your goal is to identify and follow the single best expert by observing the outcomes of the  $k$  experts at each iteration, minimizing your own loss in this process.

The exponential weights algorithm assigns each expert a score of 1, progressively down-weighting experts who make mistakes. At each iteration, the algorithm randomly picks an expert, where the probability an expert  $f \in \mathcal{F}$  is picked is proportional to  $\exp(-\sum_{i=1}^t \ell(f, t_i))$ . Informally, the algorithm prefers well-performing experts exponentially more than poor experts. For a full analysis of the algorithm and its guarantees, see [14].

We now show that the exponential weights algorithm arises naturally from a relaxation of the conditional sequential Rademacher complexity.

**Claim 1** (Proposition 3 of [3]). *Given  $\mathcal{F}$  is a finite class and  $|\ell(f, x)| \leq 1$ . Then the relaxation*

$$\text{Rel}(\mathcal{F} | y_1, \dots, y_t) = \inf_{\lambda > 0} \left\{ \frac{1}{\lambda} \log \left( \sum_{f \in \mathcal{F}} \exp \left( -\lambda \sum_{s=1}^t \ell(f, y_s) \right) \right) + 2\lambda(T - t) \right\}$$

*upper bounds  $\mathfrak{R}(\mathcal{F} | y_1, \dots, y_t)$ . Furthermore, it is an admissible relaxation with exponential weights as a corresponding algorithm.*

*Proof.* We first show the above relaxation upper bounds  $\mathfrak{R}(\mathcal{F} | y_1, \dots, y_t)$ . Again, denote  $L_t(f) = \sum_{s=1}^t \ell(f, y_s)$  as the retroactive loss of playing  $f$ . By Jensen's,  $\mathbb{E}X \leq \log \mathbb{E} \exp(X)$ , so

$$\mathbb{E}_{\epsilon} \left[ \sup_{f_t \in \mathcal{F}} 2 \sum_{s=t}^T \epsilon_s \ell(f_t, y'_{s-t}(\epsilon_{t:s})) - L_{t-1}(f_t) \right] \leq \frac{1}{\lambda} \log \mathbb{E}_{\epsilon} \exp \left( \sup_{f_t} 2\lambda \sum_{s=t}^T \epsilon_s \ell(f_t, y'_{s-t}(\epsilon_{t:s})) - \lambda L_{t-1}(f_t) \right)$$

As  $e^x$  is increasing in its argument, we move the sup out of the exponent and take a loose bound to get

$$\begin{aligned}
& \frac{1}{\lambda} \log \mathbb{E}_\epsilon \sup_{f_t} \exp \left( 2\lambda \sum_{s=t}^T \epsilon_s \ell(f_t, y'_{s-t}(\epsilon_{t:s})) - \lambda L_{t-1}(f_t) \right) \\
& \leq \frac{1}{\lambda} \log \mathbb{E}_\epsilon \sum_{f_t \in \mathcal{F}} \exp \left( 2\lambda \sum_{s=t}^T \epsilon_s \ell(f_t, y'_{s-t}(\epsilon_{t:s})) - \lambda L_{t-1}(f_t) \right) \\
& \leq \frac{1}{\lambda} \log \left[ \sum_{f_t \in \mathcal{F}} \exp(-\lambda L_{t-1}(f_t)) \cdot \mathbb{E}_\epsilon \exp \left( 2\lambda \sum_{s=t}^T \epsilon_s \ell(f_t, y'_{s-t}(\epsilon_{t:s})) \right) \right]
\end{aligned}$$

By Claim 2,  $\mathbb{E}_\epsilon \exp \left( 2\lambda \sum_{s=t}^T \epsilon_s \ell(f_t, y'_{s-t}(\epsilon_{t:s})) \right) \leq \exp(2\lambda^2 \max_\epsilon \sum_{s=t}^T \ell(f_t, y'_{s-t}(\epsilon_{t:s}))^2)$ . As each  $\ell(\cdot, \cdot)$  is bounded, this is at most  $\exp(2\lambda(T-t))$ , so the above is bounded above by

$$\frac{1}{\lambda} \log \left[ \sum_{f_t \in \mathcal{F}} \exp(-\lambda L_{t-1}(f_t)) \right] + 2\lambda(T-t)$$

As this holds for any  $\lambda > 0$ , we conclude that the above relaxation upper bounds  $\mathfrak{R}(\mathcal{F}|y_1, \dots, y_t)$ .

Next, we show that it is an admissible relaxation. For  $\phi_t = \sum_{f \in \mathcal{F}} e^{-\lambda L_t(f)}$ , note that

$$\text{Rel}(\mathcal{F}|y_1, \dots, y_t) = \inf_{f_t} \sup_{y_t} \left[ \mathbb{E}[\ell(f_t, y_t)] + \frac{1}{\lambda} \log \left( \sum_{f \in \mathcal{F}} e^{-\lambda L_t(f)} \right) + 2\lambda(T-t) \right]$$

Observe that  $\frac{1}{\lambda} \log \left( \sum_{f \in \mathcal{F}} e^{-\lambda L_t(f)} \right) = \frac{1}{\lambda} \log \phi_t$ . For  $f_t \sim \mathcal{F}$  sampled by the exponential weights distribution,  $\Pr(f_t = f) = \exp(-\lambda L_{t-1}(f)) / \phi(t-1)$ . Then,

$$\begin{aligned}
\mathbb{E}_{q_t} \exp(-\lambda \ell(f, y_t)) &= \frac{1}{\phi_{t-1}} \sum_{f \in \mathcal{F}} \exp \left( -\lambda \sum_{s=1}^{t-1} \ell(f, y_s) \right) \exp(-\lambda \ell(f, y_t)) \\
&= \frac{\phi_t}{\phi_{t-1}}
\end{aligned}$$

and thus  $\frac{1}{\lambda} \log \phi_t = \frac{1}{\lambda} \log \phi_{t-1} + \frac{1}{\lambda} \log \mathbb{E} \exp(-\lambda \ell(f, y_t))$ . Since  $\ell(\cdot, \cdot) \in [-1, 1]$ , by Lemma A.1 of [15],  $\frac{1}{\lambda} \log \mathbb{E} \exp(-\lambda \ell(f, y_t)) \leq -\mathbb{E} \ell(f, y_t) + \lambda/2 \leq -\mathbb{E} \ell(f, y_t) + 2\lambda$ . Substituting our upper bound on  $\frac{1}{\lambda} \log \phi_t$ , we have that

$$\inf_{f_t} \sup_{y_t} \left[ \mathbb{E}[\ell(f_t, y_t)] + \frac{1}{\lambda} \log \phi_{t-1} - \mathbb{E}[\ell(f, y_t)] + 2\lambda + 2\lambda(T-t) \right]$$

$$= \frac{1}{\lambda} \log \phi_{t-1} + 2\lambda(T - t + 1) = \text{Rel}(\mathcal{F}|y_1, \dots, y_{t-1})$$

We conclude the relaxation is admissible, with exponential weights as a suitable algorithm.  $\square$

This also gives an upper bound on the performance of the exponential weights algorithm:

**Corollary 3.2.1.** *For optimal choice of  $\lambda$ , the exponential weights algorithm obtains regret bounded by  $\text{Reg}(\pi_T) \leq 2\sqrt{2T \log |\mathcal{F}|}$ .*

*Proof.* The algorithm’s regret is bounded by

$$\text{Reg}(\pi_T) \leq \mathfrak{R}(\mathcal{F}) \leq \text{Rel}(\mathcal{F}) = \frac{1}{\lambda} \log |\mathcal{F}| + 2\lambda T$$

Setting  $\lambda = \sqrt{\frac{\log |\mathcal{F}|}{2T}}$  gives us the desired bound.  $\square$

## 4 Conclusion

Any theoretical minimax value of an online learning game can be obtained by a learner willing to brute force exponentially many strategies, but of greater interest is whether efficient such algorithms exist. It is apparent that conditional value relaxations such as the sequential Rademacher complexity are useful in deriving computationally tractable approaches. In this work, we investigated such relaxations as a data-dependent measure of the learner’s ability to generalize when confronted with adversarial uncertainty. Indeed, many established algorithms arise naturally from some relaxation of the conditional sequential Rademacher complexity, as shown in this work for exponential weights.

Ultimately, the Rademacher complexity serves as a powerful tool in the realm of online learning, not just for statistical learning theory from which it originated, but also for algorithmic development. Additional work on this topic may generalize the relaxation techniques to more complex action spaces, partial information settings, and adaptive notions of complexity that evolve with the observed data.

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## 5 Appendix

**Claim 2.** For independent Rademacher random variables  $\epsilon_{1:T}$ , bounded loss functions  $\ell_t$  dependent on  $\epsilon_{1:t}$ , we have that

$$\mathbb{E}_\epsilon \exp\left(\lambda \sum_{s=t}^T \epsilon_s \ell_s(\epsilon)\right) \leq \exp\left(\frac{\lambda^2}{2} \max_{\epsilon_{t:T}} \sum_{s=t}^T \ell_s(\epsilon)^2\right)$$

*Proof.* We induct backwards on  $t$ . For  $t = T$ , we note that  $-\max_\epsilon \ell_s(\epsilon) \leq \epsilon_s \ell_s(\epsilon) \leq \max_\epsilon \ell_s(\epsilon)$ . By Hoeffding's (lemma 9.2 of [12]), we immediately get

$$\mathbb{E}(\exp \lambda \epsilon_s \ell_s(\epsilon)) \leq \exp\left(\frac{\lambda^2}{2} \max_{\epsilon_T} \ell_s(\epsilon)^2\right)$$

Next, assume true for some  $t \leq T$ . We have that

$$\mathbb{E} \exp \lambda \sum_{s=t-1}^T \epsilon_s \ell_s(\epsilon) = \mathbb{E} \exp\left(\lambda \sum_{s=t}^T \epsilon_s \ell_s(\epsilon_{1:s}) + \lambda \epsilon_{t-1} \ell_{t-1}(\epsilon_{1:t-1})\right) \quad (1)$$

$$= \mathbb{E} \left[ \exp(\lambda \epsilon_{t-1} \ell_{t-1}(\epsilon_{1:t-1})) \exp\left(\lambda \sum_{s=t}^T \epsilon_s \ell_s(\epsilon_{1:s})\right) \right] \quad (2)$$

Let  $A_t := A_t(\epsilon_{1:t-1}) = \exp(\lambda \sum_{s=t}^T \epsilon_s \ell_s(\epsilon_{1:s}))$ . We have that  $\mathbb{E}[A_t] = \mathbb{E}[A_t | \epsilon_{1:t-1}]$  and inductively,  $\mathbb{E}A_t \leq \exp(\frac{\lambda^2}{2} \max_\epsilon \sum_{s=t}^T \ell_s(\epsilon)^2)$ . Now, we condition on Rademacher r.v.'s  $\epsilon_{1:t-1}$ , which gives us

$$\begin{aligned} & \frac{1}{2} (\mathbb{E}[\exp(\lambda \ell_{t-1}(\epsilon_{1:t-1}) A_t) \mid \epsilon_{t-1} = 1] + \mathbb{E}[\exp(-\lambda \ell_{t-1}(\epsilon_{1:t-1}) A_t) \mid \epsilon_{t-1} = -1]) \\ & \leq \frac{1}{2} \left( \max_{\epsilon_{t-1}} (\exp(\lambda \ell_{t-1}(\epsilon_{1:t-1})) + \exp(-\lambda \ell_{t-1}(\epsilon_{1:t-1}))) \right) \mathbb{E}[A_t | \epsilon_{1:t-1}] \end{aligned}$$

which uses that  $\exp(\lambda \ell_{t-1}(\epsilon_{1:t-1})) \leq \max_{\epsilon_{t-1}} \exp(\lambda \ell_{t-1}(\epsilon_{1:t-1}))$ , where the latter is independent of  $\epsilon_{t-1}$  and can hence be extracted from the expectation. We are implicitly conditioning on  $\epsilon_{1:t-2}$ , which combined with the conditioning on  $\epsilon_{t-1}$  leaves us with  $\mathbb{E}[A_t | \epsilon_{1:t-1}]$ . Finally, as  $\frac{e^x + e^{-x}}{2} \leq e^{-x^2/2}$  for all  $x \in \mathbb{R}$ , the above is bounded above by

$$\max_{\epsilon} \exp\left(\frac{\lambda^2}{2} \sum_{s=t-1}^T \ell_s(\epsilon)^2\right)$$

where we merge the  $\max_{\epsilon_{t-1}}$  with the  $\max_\epsilon$  that appears when substituting the inductive hypothesis.  $\square$